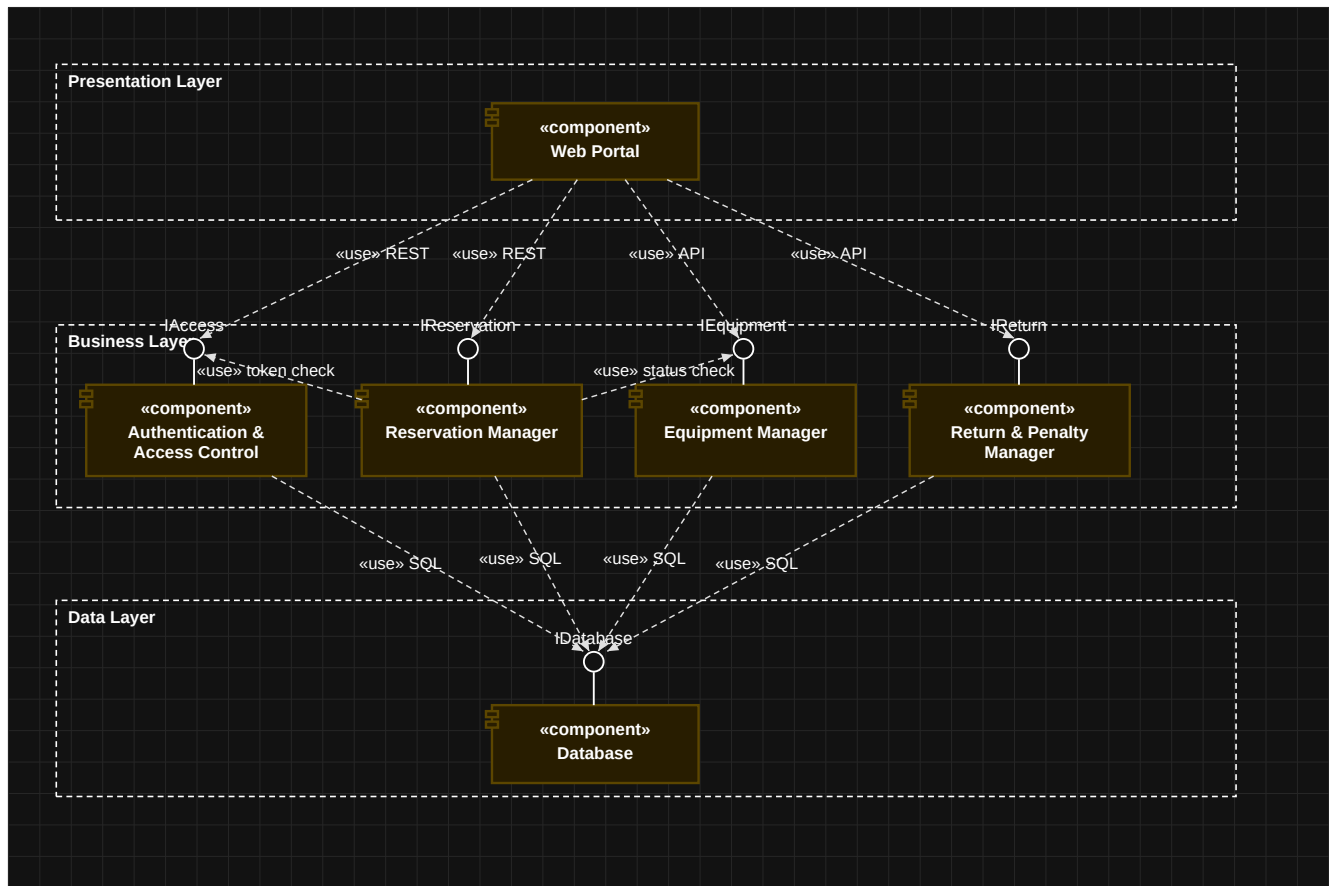


Lab 3: Component Modelling & Architectural Pattern Selection

System: Smart Lab Equipment & Slot Reservation Portal

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Architecture Selection: We chose a **Layered (N-tier) Architecture** for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice: The system is organised into three layers. The Presentation layer holds the Web Portal used by students and lab technicians. The Business layer holds four components: Authentication & Access Control, Reservation Manager, Equipment Manager, and Return & Penalty Manager. The Data layer holds the Database. Each layer calls only the layer below it, through defined interfaces (IAccess, IReservation, IEquipment, IReturn, IDatabase). Reservation Manager validates the session token through IAccess and checks equipment status through IEquipment before it books a slot.

Reason 1: Centralised slot locking prevents double-booking. Concurrent reservation requests must never book the same slot twice (FR-001, NFR-001). Only the Reservation Manager writes bookings, and it does so inside one database transaction with a row-level lock on the slot. With a single shared database this is simpler and more reliable than keeping data consistent across several independently owned services.

Reason 2: Components map directly to the requirements. Reservation and cancellation (FR-001, FR-002), equipment status (FR-003), and usage tracking with late-return penalties (FR-004, FR-005) each correspond to one business component. Each one can be built, tested and replaced on its own behind its interface. A campus lab with a small, known user base does not need the deployment and monitoring overhead of microservices.

Security Advantage: Session tokens are checked through the Authentication & Access Control component, so role-based access control (NFR-002) is enforced in one place: students reach only their own reservations, while lab technicians get equipment management and penalty functions. The Web Portal never accesses the database directly, which reduces the risk of unauthorised data access.

Performance Benefit: All business components run in the same deployment, so calls between them are in-process and add no network hops. This helps keep reservation processing within the 200 ms target (NFR-001).

Trade-off: The single database is a possible bottleneck and a single point of failure. At this lab's scale, regular backups and an indexed slot table are enough, and the layered design lets the data layer be scaled or replicated later without changing the upper layers.